

Intelligent Pipeline Condition Prediction using Machine Learning for Predictive Maintenance

- Authors: Subhash Nannapaneni, Prashanthi Thota, Zazilah Binti May
- Affiliations: SRM University, AP (India) & Universiti Teknologi PETRONAS (Malaysia)
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Introduction

- • Oil and gas pipelines are critical for global energy distribution.
- • Failures often arise from corrosion, erosion, and fatigue.
- • Traditional inspections are costly and reactive.
- • Predictive maintenance using machine learning offers proactive, cost-efficient solutions.

Research Objective

- • Develop a machine-learning-based system to classify pipeline conditions as Normal, Moderate, or Critical.
- • Benchmark multiple ML models to identify the best performer.
- • Provide a scalable, intelligent predictive maintenance framework.

Literature Review

- • ML outperforms rule-based inspection methods (Li & Zhang, 2022).
- • Gradient boosting improves predictive accuracy (Wang et al., 2023).
- • CatBoost effective for heterogeneous pipeline data (Kim & Lee, 2022).
- • Explainable AI (XAI) increases trust in ML decisions (Garcia & Patel, 2021).

System Architecture

- Data Acquisition → Data Preprocessing → Feature Engineering → Model Training → Evaluation & Interpretation
- Tools Used: Python, Pandas, Scikit-learn, Matplotlib, SHAP
- This architecture allows seamless integration of multiple algorithms and feature analysis.

Dataset Overview

- • Source: Kaggle - Predictive Maintenance for Oil & Gas Pipelines
- • Variables: Pipe Size, Thickness, Material, Grade, Pressure, Temperature, Corrosion Impact, etc.
- • Output: Condition (Normal / Moderate / Critical)
- • Data split: 80% Training, 20% Testing (Stratified)

Preprocessing & Feature Engineering

- • Handled missing values using mean/mode substitution.
- • Applied one-hot encoding for categorical variables.
- • Scaled continuous features using Min-Max normalization.
- • Removed outliers via Z-score method.
- • Identified key predictive features: Thickness Loss, Material Loss, Corrosion Impact.

Models Compared

- LightGBM – Gradient boosting with leaf-wise growth, high efficiency.
- CatBoost – Handles categorical data automatically.
- Random Forest – Baseline ensemble model, interpretable and stable.
- Extra Trees – Randomized splits for variance reduction.
- SVM (RBF) – Nonlinear classifier for small-scale patterns.

Performance Comparison

- Model | Accuracy | Precision | Recall | F1-Score
- LightGBM | 0.96 | 0.94 | 0.93 | 0.93
- CatBoost | 0.93 | 0.90 | 0.89 | 0.89
- Random Forest | 0.92 | 0.88 | 0.87 | 0.87
- Extra Trees | 0.91 | 0.87 | 0.86 | 0.86
- SVM | 0.90 | 0.86 | 0.84 | 0.85
- → LightGBM outperformed all models achieving 96% accuracy.

Key Findings

- • Gradient boosting algorithms outperform traditional ensembles.
- • LightGBM achieved highest accuracy and balanced performance.
- • SHAP interpretation revealed most influential parameters.
- • Demonstrates strong potential for industrial predictive maintenance.

Limitations

- • Dataset lacks real-time IoT sensor data.
- • Slight class imbalance affects generalization.
- • Static data – does not account for time-dependent behavior.
- • Environmental and operational variability not modeled.

Conclusion & Future Work

- Conclusion:
 - • LightGBM-based predictive model achieved superior results.
 - • Enables proactive maintenance and reduced downtime.
- Future Work:
 - • Integrate real-time IoT data for dynamic prediction.
 - • Explore hybrid boosting + deep learning methods.
 - • Expand datasets across materials and conditions.
 - • Apply Explainable AI for transparent maintenance planning.